TASK-8

- **CODE :**

```
# Advanced OOP Concepts

# Parent Class
class Person:
    def __init__(self, name):
        self.name = name

    def show(self):
        print("Name:", self.name)

# Child Class (Inheritance)
class Student(Person):
    def __init__(self, name, course):
        super().__init__(name)
        self.course = course

    # Polymorphism (Method Override)
    def show(self):
        print("Student Name:", self.name)
        print("Course:", self.course)

# Encapsulation
class BankAccount:
    def __init__(self, balance):
        self.__balance = balance

    def get_balance(self):
        return self.__balance

    def deposit(self, amount):
        self.__balance += amount
```

```
# Creating Objects
student = Student("Shru", "Python Internship")
student.show()

account = BankAccount(1000)
account.deposit(500)

print("Balance:", account.get_balance())
```

Output

```
Student Name: Shru
Course: Python Internship
Balance: 1500

=== Code Execution Successful ===
```

- **DELIVERABLES:**

1. **What is Inheritance?**

Inheritance is a feature of OOP where one class acquires properties and methods of another class.
Example:
Student inherits from Person.

2. Static vs Instance Methods
Instance Method:
Works with object data and uses self.

Static Method:
Belongs to the class and does not use self.

- **README.md:**

```
# Edutech Python Task 8

## Topic
Advanced OOP Concepts

## Files Included
- advanced_oop.py
- answers.txt
- screenshot

## Concepts Used
- Inheritance
- Polymorphism
- Encapsulation
- Classes and Objects
```